

POWIDZ, Poland

WMO: 123360

Lat: 52.379N

Lon: 17.854E

Elev: 113

StdP: 99.97

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-11.7	-17.2	0.8	-13.8	-14.5	1.1	-10.8	13.7	6.7	12.0	4.8	2.5	60	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	30.9	19.4	29.0	18.7	27.2	18.1	20.9	27.6	20.1	26.5	19.2	24.9	4.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.9	22.3	18.0	13.1	21.9	17.1	12.4	21.3	60.6	27.6	57.9	26.9	55.2	24.8	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
10.2	8.9	7.9	DB	-17.8	34.2	4.1	1.5	-20.7	35.3	-23.1	36.1	-25.4	37.0	-28.4	38.0	
			WB	-17.6	22.4	4.3	1.0	-20.7	23.1	-23.2	23.7	-25.7	24.3	-28.8	25.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-1.8	-0.2	3.4	9.3	13.9	17.4	19.8	19.2	14.4	9.1	4.5	0.7
	DBStd	8.51	5.48	4.98	4.52	4.23	4.11	3.55	3.35	3.18	3.32	3.92	3.54	4.59
	HDD10.0	1452	366	284	206	62	11	0	0	0	3	64	167	289
	HDD18.3	3503	624	518	462	271	147	59	23	27	124	285	415	548
	CDD10.0	1162	0	0	3	42	132	222	302	287	136	38	1	0
	CDD18.3	172	0	0	0	1	11	31	67	56	7	0	0	0
	CDH23.3	1669	0	0	0	18	116	324	640	494	76	1	0	0
	CDH26.7	495	0	0	0	3	22	91	212	147	19	0	0	0
Wind	WSAvg	3.7	4.2	4.0	4.0	3.7	3.5	3.5	3.5	3.3	3.4	3.6	3.9	4.2
Precipitation	PrecAvg	528	33	27	35	30	53	60	84	57	44	35	35	35
	PrecMax	717	68	59	68	78	103	114	152	161	100	116	103	80
	PrecMin	350	4	4	9	8	19	19	19	16	16	6	1	7
	PrecStd	85	17	12	16	16	22	27	35	32	23	24	18	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	11.6	17.3	25.7	29.1	32.5	33.4	33.1	29.3	21.9	14.1	10.9
		MCWB	8.5	8.5	10.7	14.8	18.0	20.6	20.0	20.9	18.6	16.0	11.3	9.1
	2%	DB	8.3	9.5	14.6	22.0	26.5	29.1	31.1	30.2	25.4	19.1	12.1	9.0
		MCWB	6.6	7.2	8.9	13.5	16.9	18.8	19.2	19.3	16.9	14.3	10.3	7.8
	5%	DB	6.6	7.7	12.2	19.3	24.1	27.0	29.1	28.1	23.0	17.0	10.6	7.8
		MCWB	5.4	5.7	8.1	11.7	15.9	17.7	18.7	18.6	15.8	13.3	9.3	6.7
	10%	DB	5.1	6.1	10.2	17.0	21.8	24.8	27.0	26.1	20.9	14.9	9.5	6.3
		MCWB	4.0	4.3	7.1	10.8	14.7	16.7	18.2	17.8	14.9	12.4	8.3	5.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	9.2	11.8	15.9	19.5	21.9	21.8	21.8	19.3	16.5	12.0	9.4
		MCDB	10.2	10.7	16.1	22.9	26.3	29.5	27.7	28.9	27.3	20.4	13.4	10.2
	2%	WB	6.9	7.4	9.9	14.1	17.9	20.1	20.9	20.6	17.7	14.8	10.6	8.0
		MCDB	8.3	9.0	13.0	20.9	24.7	26.8	27.5	27.3	23.2	18.0	11.7	8.7
	5%	WB	5.5	5.8	8.6	12.6	16.7	18.9	20.1	19.8	16.6	13.7	9.4	6.7
		MCDB	6.6	7.6	11.7	18.0	22.4	24.5	26.4	26.0	21.3	16.4	10.4	7.6
	10%	WB	4.0	4.4	7.3	11.3	15.5	17.7	19.3	18.9	15.6	12.6	8.3	5.5
		MCDB	5.1	5.9	10.0	15.9	20.4	22.7	24.9	23.8	19.5	14.8	9.3	6.3
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.8	7.9	10.5	10.7	10.9	10.6	10.7	9.6	7.7	5.0	4.5
		MCDBR	4.9	7.5	11.7	14.0	13.8	14.2	13.9	13.8	12.9	10.6	6.8	5.2
		MCWBR	4.2	5.7	7.4	7.2	6.0	6.0	4.6	5.3	5.9	6.3	5.2	4.5
	5% WB	MCDBR	5.0	7.4	10.5	12.6	12.2	12.2	11.5	11.5	11.2	9.6	6.6	5.1
		MCWBR	4.4	5.9	7.1	6.8	5.9	6.0	4.7	4.9	5.7	6.1	5.3	4.5
Clear-Sky Solar Irradiance	taub	0.314	0.341	0.376	0.421	0.416	0.410	0.439	0.429	0.389	0.367	0.344	0.310	
	taud	2.282	2.282	2.234	2.161	2.228	2.274	2.208	2.241	2.325	2.363	2.336	2.313	
	Ebn at Noon	666	760	804	810	836	844	808	795	784	722	624	604	
	Edh at Noon	64	86	110	134	131	127	133	122	100	80	62	53	
All-Sky Solar Radiation	RadAvg	0.67	1.35	2.54	4.15	5.14	5.38	5.00	4.42	3.18	1.75	0.77	0.49	
	RadStd	0.11	0.23	0.33	0.54	0.56	0.48	0.60	0.39	0.41	0.25	0.10	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	-156	+71	N/A

Nomenclature: See separate page